

Statement of Work Optoelectronics, Power Electronics, and Radiation Effects Research

1.0 INTRODUCTION

The Electronic Sciences and Technology Division (ESTD) of the U.S. Naval Research Laboratory is tasked with producing novel electronic devices having advanced properties and capabilities, and transitioning these devices to the warfighter. This is achieved through coordinated modeling, simulation, and design efforts. Designs are realized in working devices through semiconductor growth and fabrication, and the device performance is measured in the characterization laboratories within ESTD. ESTD researches technologies ranging from basic electronic materials to high-powered electronics, high-speed microwave devices, and optoelectronics, each with their own critical performance parameters such as efficiency of operation, responsiveness, and power output.

Within ESTD, Codes 6810 and 6880 are engaged in the development of advanced semiconductor materials, devices, and circuits for use in a wide array of applications including photovoltaics, infrared imaging, and advanced materials and devices for space applications. The potential application areas for these new technologies address a broad range of ground and space-based military missions, so the research includes reliability testing, physics of failure analysis, and space environmental testing. The goal of this task is to provide scientific and engineering support for advanced semiconductor materials, devices, and circuits to increase capability and reliability of military missions for both terrestrial and space applications.

2.0 OBJECTIVE

With such a wide range of research, there are often gaps in expertise for specific technical areas. The objective of this work is to provide Research and Development (R&D) capability to Codes 6810 and 6880 in key technical gap areas. This work will bring technical versatility, and agility to Codes 6810 and 6880 by enabling access to technical skills in a variety of areas. By so doing, we will enable continued responsiveness to the warfighter.

3.0 SCOPE

The contractor shall provide expertise to assist in a wide range of NRL research activities related to optoelectronics, power electronics, and radiation effects. The following task areas and descriptions indicate the specific nature of these research efforts.

4.0 TASKS

All tasks and sub-tasks are subject to CDRL A001, “Technical Reports & Presentation Materials”.

4.1 Task 1: Semiconductor Material/Device/Circuit Modeling, Simulation, and Design

4.1.1. The Contractor shall perform theoretical research to develop physics-based numerical models of electronic and atomic structure of electronic and optoelectronic devices. The goal of these efforts is to develop a theoretical basis for understanding the physics of operation of these systems.

4.1.2. The Contractor shall conduct experimental research on device model execution and interpretation of experimental data for device operation.

4.1.3. The Contractor shall develop specialized physics-based computer models for the

simulation of linear and nonlinear optical interactions of ultrashort optical pulses with semiconductor materials and devices, and apply those models to understanding carrier generation and transport in materials and devices.

4.2 Task 2: Semiconductor Growth, Processing and Device Fabrication

4.2.1. The Contractor shall grow semiconductor materials for producing the devices modeled in Task 4.1. The growth techniques shall include molecular beam epitaxy (MBE), organometallic vapor phase epitaxy, (OMVPE), atomic layer deposition (ALD), and sputter deposition as well as those techniques specified by the Task Manager.

4.2.2. The Contractor shall fabricate devices from materials grown in 4.2.1. The goal of this effort is to produce functional devices such as optical detectors, solar cells, field-effect transistors (FETs), and diodes that can be electrically or optoelectronically measured.

4.2.3. The Contractor shall fabricate integrated circuits by using the devices developed in 4.2.2, and integrating commercial off-the-shelf (COTS) parts for applications to be specified by the Task Manager. The application of interest often requires development beyond the discrete part (e.g. an integrated circuit). In these cases, circuits are to be fabricated using the parts that have been developed, and COTS as appropriate. Examples include optoelectronic applications such as focal plane arrays and the associated read-out circuits and solar cell arrays.

4.2.4. The Contractor shall fabricate and process semiconductor devices including the following: Removal of residual oxides from the backs of silicon wafers using both wet chemistry and plasma processes; deposition of metals using both shadow and lithographic masks, including the use of special non-radiative deposition equipment; performs material characterization using spectroscopic ellipsometry, stylus profilometry, and other characterization techniques as required for the specified material system.

4.3 Task 3: Semiconductor Material and Device Characterization

4.3.1. The Contractor shall characterize structural attributes for the materials, devices, and circuits grown and fabricated under this project. Structural characterization techniques may include optical microscopy, ellipsometry, x-ray diffraction, and electron microscopy. The results of these characterizations determine the atomic and crystalline structure of the materials grown within the project.

4.3.2. The Contractor shall characterize electrical attributes for the materials, devices, and circuits that are grown and fabricated under this project. Electrical characterization techniques include current-voltage (I-V), capacitance-voltage (C-V), AC conductance, and Hall effect. Characterization of active and/or passive microwave devices involves use of network analyzers, spectrum analyzers and other microwave test equipment to determine the linear and/or nonlinear characteristics of the devices.

4.4 Task 4: Reliability and Environmental Testing of Semiconductor Materials and Devices in Simulated Space Environments.

4.4.1. The goal of this task is to determine how the devices and circuits of interest perform in operational environments.

4.4.2. The Contractor shall conduct reliability testing for devices studied in this project. The reliability testing refers to measurement of the pertinent parameters from the devices and circuits to include electrical, optical and/or thermal properties of the devices. The reliability testing also includes failure analysis of the failed devices to determine failure modes and mechanisms.

4.4.3. The Contractor shall perform space environmental testing to evaluate the effects of particle irradiation on satellite subsystems, including, but not limited to: low-energy proton testing, energetic particle testing of optoelectronics and imaging systems, and

heavy-ion and proton evaluations of single particle interactions in devices and integrated circuits, including high-speed technologies operating at gigabit per second data rates. Environmental testing is done to demonstrate the device and/or circuit performance in a relevant environment. This testing includes device and circuit characterization before, during, and after exposure to various environmental factors including electrical bias, particle irradiation, and optical approaches to simulating particle radiation effects. These measurements also include annealing studies after the simulated space environmental exposure.

4.4.3. The Contractor shall develop and implement linear and nonlinear optical approaches for carrier injection into semiconductor materials and devices for the purposes of simulating the effects of particle radiation on the materials and devices. This task includes laser source development, experimental optical beam line development and characterization, measurement of the linear and nonlinear optical properties of relevant materials and material systems, as well as linear and nonlinear-optical experimental design and implementation for the purposes simulating the effects of particle irradiation on the materials, devices, and circuits of interest.

4.4.4. The Contractor shall perform linear and nonlinear optical measurements to simulate the effects of particle irradiation on semiconductor devices and circuits. This task shall include sample preparation and mounting, test setup design and implementation, including board design for both static and high-speed measurements, optical system characterization and calibration, and data acquisition and analysis.

4.4.5. The Contractor shall perform linear and nonlinear optical measurements on materials of interest to this effort, as appropriate, including, but not limited the nonlinear-optical z-scan measurements, linear and nonlinear absorption, stimulated Raman gain, four-wave mixing, and pump-probe measurements.

4.5 Task 5: High Power Electronics

4.5.1. Power Electronic Circuits: The Contractor shall develop high performance, high efficiency power conversion circuits (including wireless power transfer) enabled by and implementing relevant wide bandgap power electronic materials and devices.

4.5.2. Modeling and Simulation: The Contractor shall develop semiconductor device models, utilize available simulation code, and apply each to novel wide bandgap semiconductor device structures.

4.5.3. Advanced Wide Bandgap Deposition and Characterization: The Contractor shall utilize advanced deposition techniques and optical/electrical techniques to characterize wide bandgap semiconductor materials and devices.

4.5.3.1: The Contractor shall utilize transient optical characterization techniques to characterize the properties of wide bandgap semiconductor materials and device.

4.5.3.2: The Contractor shall use Fourier Transform optical characterization techniques to characterize the properties of wide bandgap semiconductors.

4.5.3.3: The Contractor shall develop custom control software and hardware design and construction in support of wide bandgap semiconductor characterization efforts.

4.5.4: Wide Bandgap Semiconductor Device Fabrication: The Contractor shall use semiconductor deposition and processing techniques to fabricate a wide variety of wide bandgap semiconductor devices.

NOTE: The following Task descriptions 4.6 to 4.8 are unclassified, but the results will be **CLASSIFIED AT A TS/SCI LEVEL. ALL RESULTS MUST BE DELIVERED THROUGH SECURE CHANNELS.**

4.6 Task 6. Semiconductor Device/Circuit Modeling, Simulation, and Design

4.6.1. The Contractor shall perform theoretical research to develop numerical models of electronic and atomic structures for optoelectronic, microwave, and power electronic devices. The goal of these efforts is to create a theoretical basis for the design of novel materials for advanced electronic devices.

4.6.2. The Contractor shall conduct experimental research on device model execution and interpretation of experimental data for device operation in its intended environment. Analyses are intended to produce a model for the operational environment and simulations and predictions of the device and circuit operation in the modeled environment. The results of these efforts will serve to validate the models produced in **4.6.1.** and provide insight into means for improving the models and new technological paths to pursue for new devices.

4.6.3. The Contractor shall develop specialized physics-based computer models for design, modeling, and simulation of novel materials and devices. The description of the novel materials and devices developed in this research may often require the development of new models based on the physical and engineering principles operative in these materials and devices.

4.7 Task 7. Semiconductor Growth and Device/Circuit Fabrication

4.7.1. The Contractor shall grow semiconductor materials for producing the devices modeled in task 4.6. Growth techniques may include MBE, OMVPE, ALD and sputter deposition as well as those specified by the Task Manager. **Deliverables:** Semiconductor materials grown for fabrication, [CDRL A002](#).

4.7.2. The Contractor shall fabricate devices modeled in Task 4.6 to produce functional devices such as optical detectors, solar cells, FETs, and diodes. **Deliverables:** Devices fabricated from materials in Task 4.6.1, [CDRL A003](#).

4.7.3. The Contractor shall fabricate integrated circuits by using the parts/devices of task 4.7.2 and integrating COTS parts for applications to be specified by the Task Manager. Examples include focal plane arrays and the associated read-out circuits, solar cell arrays, analog circuits for power conversion and switching devices, and waveguide and microstrip circuits for microwave applications. **Deliverables:** Circuits fabricated from devices in Task 4.7.3, [CDRL A004](#).

4.8 Task 8. Semiconductor Material and Device Characterization in Simulated Environments

4.8.1. The Contractor shall characterize the structural attributes of the materials, devices, and circuits of Task 4.7. Structural characterization techniques may include optical microscopy, ellipsometry, x-ray diffraction, and electron microscopy. The results of these characterizations determine the atomic and crystalline structure of the materials grown within the project.

4.8.2. The Contractor shall characterize electrical attributes for the materials, devices, and circuits that are grown and fabricated under this project. Electrical characterization techniques include current-voltage (I-V), capacitance-voltage (C-V), AC conductance, and Hall effect. Characterization of active and/or passive microwave devices involves use of network analyzers, spectrum analyzers and other microwave test equipment to determine the linear and/or nonlinear characteristics of the devices.

4.8.3. The Contractor shall characterize optoelectronic attributes for the materials, devices, and circuits that are grown and fabricated under this project. Optoelectronic measurements include quantum efficiency, electroluminescence (EL), Fourier Transform Infrared Spectroscopy (FTIR), and I-V measurements under simulated solar illumination. The choice of specific technique depends on the material, device, or circuit under study.

5.0 PERIOD OF PERFORMANCE

The period of performance is expected to be one base year with 4 one-year option periods.